



U.S. Department
of Transportation
MARITIME
ADMINISTRATION

1200 New Jersey Ave, S.E.
Washington, D.C. 20590

**A Study on the Economic Impact of the
International Maritime Organization Net Zero Framework
RFP No. 693JF726R000015
05/04/2026**

REQUEST FOR PROPOSALS (RFP):

The U.S. Maritime Administration is commissioning a targeted cost analysis of the International Maritime Organization's (IMO) "Net Zero Framework" (NZF), adopted in April 2025, that centers on the NZF's specific policy mechanisms and their economic consequences **for the United States**. The study should examine, at minimum, the NZF's:

- (a) global GHG pricing/levy proposals (including scope, price trajectory, and pass-through mechanisms).
- (b) emissions trading and crediting systems.
- (c) use and role of carbon offsets and related crediting rules.
- (d) fuel/energy standard or fuel blending mandates.
- (e) technology mandates, exemptions, and phase in schedules.
- (f) monitoring, reporting, and verification, auditing and compliance regimes.
- (g) Fund governance, revenue collection, and allocation rules (including conditionality and funding flows to ports, projects, or countries); and
- (h) enforcement, penalties, and potential border or trade adjustment measures.

MARAD is particularly interested in how each mechanism could:

- (1) alter operating and capital costs for U.S. flagged and U.S. dependent shipping and logistics;
- (2) affect import/export unit costs (TEU/ton) and trade competitiveness.
- (3) change demand for U.S. energy, technology, and maritime services.
- (4) create administrative and compliance burdens; and
- (5) risk leakage, double counting, or ineffective mitigation (for example, reliance on offsets that do not deliver measurable reductions).

The study should quantify impacts where feasible, identify key assumptions and uncertainties for each mechanism, and analyze distributional effects across U.S. industries, ports, and regions.

The contractor should compare outcomes under: (A) the NZF as proposed (mechanism by mechanism); (B) alternative designs that limit or reshape levy/offset components; and (C) delayed or phased implementation scenarios. For each mechanism, the study must recommend practical safeguards, alternative policy instruments, or design changes that would reduce unnecessary economic burdens on U.S. interests while maintaining emissions outcomes rooted in reality, and provide clear metrics and analytical methods that MARAD and U.S. negotiators can use to evaluate NZF proposals moving forward. For each mechanism, the study shall compare

the proposed mechanism relative to a “No Action” approach to fully distinguish the costs and benefits to U.S. maritime, trade, and economic interests.

Project must be completed within a 3-month timeline.

Maximum MARAD funding available for this project is \$280,000

A. PROGRAM DESCRIPTION:

MARAD is investigating the costs and other economic considerations associated with emerging international regulations at the International Maritime Organization (IMO). Since the IMO “2023 Revised GHG Strategy” was implemented, the forum committed to developing “mid-term measures” that would include technical and economic elements. Negotiations following the revised strategy culminated in April 2025 with the IMO Maritime Environmental Protection Committee’s (MEPC) approval of a “Net-Zero Framework (NZF).” The NZF would include a range of economic instruments matched to a prescribed steady decrease in the GHG intensity of energy that internationally trading vessels use, resulting in “net-zero emissions” from the industry by 2050. The U.S. has concerns about potential impacts on the U.S. maritime sector and economy should the NZF be approved.

MARAD is commissioning an economic analysis of the International Maritime Organization’s (IMO) ‘Net-Zero Framework’ (NZF), adopted in April 2025, focusing on the NZF’s specific policy mechanisms and their economic consequences for the United States. The study shall examine, at minimum, the NZF’s:

- (a) global GHG pricing/levy proposals (including scope, price trajectory, and pass-through mechanisms).
- (b) emissions trading and crediting systems.
- (c) use and role of carbon offsets and related crediting rules.
- (d) fuel/energy standard or fuel-blending mandates.
- (e) technology mandates, exemptions, and phase-in schedules.
- (f) monitoring, reporting, and verification, auditing, and compliance regimes.
- (g) Fund governance, revenue collection, and allocation rules (including conditionality and funding flows to ports, projects, or countries).
- (h) economic cost analysis across key U.S. industries, major ports, and regions (direct and indirect employment, capital expenditures, and GDP impacts) and sectoral or commodity-specific effects, consistent with the U.S. Government’s stated energy and natural resource development priorities; and
- (i) enforcement, penalties, and potential border or trade-adjustment measures.

B. Requested Study Content

The objective of this economic analysis project is for a selected contractor to consider the range of proposed and possible international regulations, agreements, and regional schemes to assess the impacts on U.S. economic interests over the next 10 years. This work will produce a narrative final report that will inform negotiations and decision making in a rapidly changing policy environment. Where practical, the report will provide quantitative analysis and outputs.

It is understood that some elements of analysis may not be quantified nor monetized and in those cases the contractor will provide objective qualitative analysis. Timing for interim outputs may be adjusted during the project to align with major meetings or other events in the regulatory development process.

Project Description and Tasks

The following sections outline four key tasks with specific content and interim deliverables for this project. Because of the rapidly changing landscape for this subject, the methods and outputs may be refined during the project period based on data availability, feedback from stakeholders, emerging new policy frameworks, and stated U.S. Government policy priorities, with any significant changes approved by MARAD's project manager.

Task 1: Context and Background

- a. Provide a summary of existing analyses specifically related to economic effects of the proposed NZF and alternatives on U.S. maritime and economic interests.
- b. Review literature with references and findings from any existing studies or claims about the broader economics of NZF or similar theoretical international frameworks; illustrate the range of estimated costs and benefits indicated in these studies, noting if any specifically indicate impacts on the U.S.
- c. Provide an overview of existing "adopted" version of the IMO NZF (from April 2025, including materials developed in subsequent working groups). Include a description of each NZF mechanism, including its intended function, regulatory scope, and relevant international negotiation history.
- d. An inventory of current known and theoretical proposals and potential alternatives to the NZF. Including a "No Action" approach. The inventory should include a breakdown of key constituencies among international country blocks and industry interests. Focus specifically on any outputs from MEPC 84 that may represent convergence on an alternative framework to the NZF and why an alternative framework may be inconsistent with U.S. economic interests.
- e. Provide a description of U.S. economic interests that could be affected both directly and indirectly by the NZF, including:
 - U.S. commercial vessels, domestic and international
 - U.S. energy suppliers.
 - U.S. technology manufacturers; and
 - U.S. imports and exports of goods and commodities (on both foreign and domestic vessels), referencing the 2024 IMO "Comprehensive Impact Assessment" as appropriate. Highlight shipped goods and commodities that have more or less "elasticity" in fluctuating vs. permanent changes to shipping costs.
- f. Provide an overview of types and magnitude of current U.S. investments that would be affected by the NZF or similar regulation.

Task 2: Economic analyses

- a. Evaluate the range of direct economic effects to:
 - U.S. flagged, owned, or operated ships; and
 - U.S. imports and exports in aggregate and per/unit (TEU/Ton) of product, using U.S. State Department figures and methods as a reference. Highlight several specific case studies using specific products being shipped (using U.S. freight examples from the 2024 IMO “Comprehensive Impact Assessment (CIA)” if possible).
- b. Evaluate the range of potential indirect economic effects:
 - To U.S. industry broadly, including technology and vessel manufacturing, fuel transportation, and infrastructure.
 - To U.S. energy suppliers specifically, both traditional energy, bio-fuels, and any other energy sources considered likely candidates for future maritime power. Highlight the potential effects on U.S. agricultural interests and potential to provide energy or technology to foreign markets. Include consideration of effects on manufacturers and downstream storage providers if uptake of carbon capture and storage systems are widely adopted to support continued use of conventional fuels. This discussion may account for the feasibility of carbon capture and storage given past performance and the realistic trajectory of such technology; and
 - Examine potential benefits or costs of Fund revenue for the U.S., including strategies for allocations that could benefit or harm U.S. interests (i.e. if U.S. infrastructure companies would benefit from building fuel infrastructure in developing countries even if that money is technically going to the other country).
- c. Comparisons:
 - For each alternative (to the NZF) proposal or scenario, including a “No Action” approach, develop a objective assessment of positive and negative net effects on each major U.S. economic interests described in Task 1d;
 - Compare the likely overall net effects of the NZF and similar schemes for the U.S. versus other key economic competitors (EU & China); and
 - Compare the economic effects of implementation of the NZF as proposed at IMO versus 1- to 5-year delays and a “No Action” approach, assuming that the 2023 IMO GHG Strategy remains the guiding principle for international regulation.

Task 3: Specific Applications

- a. Coordinating with MARAD, identify specific affected U.S. industries that require a more detailed economic analysis. As appropriate for the analysis, this may include direct engagement, case studies, published policies, and plans; and

- b. Evaluate compatibility or conflicts with goals and principles in 2026 U.S. Maritime Action Plan¹, noting potential effects on demand for additional U.S.-built vessels or technologies.

Task 4: Final Report - Summary, Review, and Recommendations

The final report will be a comprehensive, narrative report that includes all analyses, references, methods, results, and conclusions. The content of the report must be synthesized to provide both industry and policy audiences with a clear understanding of the complete economic impacts of potential regulations. The narrative report must be easy to navigate, well written, and incorporate visually compelling graphics that summarize complex topics in creative and effective ways. It must include the following elements and qualities:

- a. Discussion of topics and variables with greatest uncertainty.
- b. If practical, a quantitative cost impact by significant NZF mechanism. Otherwise, a qualitative assessment.
- c. Assessments of any broader economic or technical effects that are less quantifiable, too uncertain, or too long-term to be included in this study; and
- d. Recommendations for additional studies and future updates based on information that may become available in coming years.

All required materials must be submitted in final form within 3 months from time of award.

Proposals may include additional recommended analyses and content that the proponent believes essential to achieving the broader goals of the study.

C. FEDERAL AWARD INFORMATION:

MARAD will provide funding through a cooperative agreement for the execution of this study. This announcement is an expression of interest only and does not commit MARAD to make any award.

To be considered for selection, the contractor performing the study must be able to complete it within 3 months from time of award. The final report and other summary materials will be publicly available and posted to the META webpage.

Proposals in response to this RFP are limited to 12 pages and must be received within 20 days of the publication of this notice. Proposals will be screened for use of Artificial Intelligence (AI) tools. Any use of AI tools in generating content for the proposal must be clearly identified and justified.

MARAD reserves the right to fund one or none of the proposals received under this RFP. MARAD is not responsible for reimbursing the costs that applicants incur to prepare and submit their proposals in response to this RFP. Technical and cost proposals (or any other material) submitted in response to this RFP will not be returned. It is MARAD's policy to treat all proposals submitted under this RFP as sensitive competitive information and to disclose their contents only for the

¹ <https://www.whitehouse.gov/wp-content/uploads/2026/02/Restoring-Americas-Maritime-Dominance.pdf>

purposes of evaluation. All applicants must be registered and current with SAM.gov at the time of award.

D. ELIGIBILITY INFORMATION.

1. Eligible Applicants: Open to all types of domestic applicants other than individuals.
2. MARAD is seeking to provide full funding through cooperative agreement for the proposal that provides sufficient, verifiable, and detailed data on how to reach the objective of this feasibility study. MARAD will fund 100% of the total cost of the project. Proposals must include a detailed cost breakdown that shows how Federal funds will be used for the project. Please see Section E.2 for information on proposal format requirements.
3. Systems for Award Management (SAM) Requirements. MARAD will not make an award to an applicant until the applicant has complied with all applicable unique entity identifier and SAM requirements. If an applicant has not fully complied with the requirements by the time MARAD is ready to make an award, MARAD may determine that the applicant is not qualified to receive an award and use that determination as a basis for making an award to another applicant. Please see Section E.3 for information on SAM requirements. MARAD will not make an award to any applicant with an exclusion in SAM.

E. APPLICATION AND SUBMISSION INFORMATION

1. Address to Request Application Package.

This RFP contains all the information needed to submit a proposal and be considered for award. No specific application form or other materials need to be obtained from MARAD.

2. Content and Form of Application.

Proposals are limited to 12 pages and must be received within 20 days of the publication of this notice.

Proposals must provide sufficient, verifiable, and detailed information covering all elements of the “Required Study Content” described in section B.1 above.

3. Unique Entity Identifier and System for Award Management (SAM).

Each applicant is required to: (a) be registered and current in SAM before submitting its application; (b) provide a valid unique entity identifier in its applicant; and (c) continue to maintain an active SAM registration with current information at all times during which it has an active Federal award or an application or plan under consideration by a Federal awarding agency. MARAD will not make an award to an applicant until the applicant has complied with all applicable unique entity identifier and SAM requirements. If an applicant has not fully complied with the requirements by the time MARAD is ready to make an award, MARAD may determine that the applicant is not qualified to receive an award and use that determination as a basis for making an award to another applicant.

4. Submission Dates and Times.

The deadline for proposal submission is 12:00 PM EDT on Wednesday, June 3, 2026.

Please submit proposals in PDF format via email to: Christian Onwudiegwu at Christian.onwudiegwu@dot.gov and Kelly Mitchell-Carroll at K.mitchell-carroll@dot.gov.

Do not send paper copies, or other media of the proposal via post office or delivery service.

Proposals received by MARAD after the deadline will not be considered for award. An email will be deemed “received” by MARAD on the date and time the email was “sent” to the email address in Section E.7, below, as determined by MARAD’s servers.

5. Intergovernmental Review.

This funding opportunity is not subject to Executive Order 12372, “Intergovernmental Review of Federal Programs.”

6. Funding Restrictions.

Applicants are advised to review 2 CFR Part 200, Uniform Administrative Requirements, Cost Principles and Audit Requirements for Federal Awards as supplemented by 2 CFR Part 1201 for cost principles used to determine allowable costs of work performed under Federal awards.

7. Other Submission Requirements.

Proposals must be submitted by the deadline set forth in Section E.4 above by electronic mail to **Christian Onwudiegwu** at Christian.onwudiegwu@dot.gov and **Kelly Mitchell-Carroll** at K.mitchell-carroll@dot.gov. Please include the Funding Opportunity Number and Title in the Subject Line.

F. APPLICATION REVIEW INFORMATION

1. Review and Selection Process.

MARAD will set up an internal review team to evaluate all proposals that are submitted under this announcement. All eligible proposals will be evaluated within two weeks after the closing date. If the review team has any questions on an individual proposal, it will notify the applicant for clarification.

The proposals will be evaluated against the following criteria:

- 1) The total fixed cost proposed for the project includes a detailed and defensible budget for each component of the project, and description of any other value-added components, resources, or co-funding.
- 2) The applicant’s understanding of the overall project, its individual components, U.S. Government priorities as it relates to this project, and the economic consequences of major changes to international policy and regulation on the national economy.
- 3) The applicant organization’s technical capabilities and resources relevant to the project, including related experience, techniques, or unique combinations of these that are integral factors for achieving the study objectives. This will include consideration of the

experience and capability of key personnel involved, the number of hours budgeted for key people, and the role, value, and level of participation of any noted partners.

- 4) Well-documented and context-appropriate methodology to undertake and successfully complete each of the study components in a timely manner. Methodologies must include descriptions of major analytical approaches being proposed for the study components, including those for evaluating current and future economic conditions. Methodology also includes any specific models and data sources intended to be used for each component of the project.
- 5) Based on past work and descriptions in the proposal, the applicant's ability to deliver a product that is:
 - a. Sufficiently detailed to fully describe the information in each task.
 - b. Well synthesized, so information and analysis from each task informs other tasks and contributes to a well-organized overall study.
 - c. Applicable to the indicated subject (comprehensive economic impacts of international regulations); and
 - d. Easily useable and accessible by, and relevant to the intended audiences (U.S. policy makers).

G. FEDERAL AWARD ADMINISTRATION INFORMATION

1. Federal Award Notices.

The successful applicant(s) will be notified by MARAD's Office of Acquisition via telephone; however, the telephone call is not the actual award or the authorization to begin performance. All awards will be made by a cooperative agreement signed by MARAD's Agreement/Grants Officer. After an award has been made, a representative from the MARAD's Office of Innovation will schedule a kick-off meeting for the project.

Unsuccessful applicants will be notified via email.

2. Administrative and National Policy Requirements.

Applicants are advised to review 2 CFR Part 200, Uniform Administrative Requirements, Costs Principles and Audit Requirements for Federal Awards as supplemented by 2 CFR Part 1201, for applicable administrative and national policy requirements. For a copy of the general terms and conditions of the Federal award made under this RFP, please contact **Kelly Mitchell-Carroll, Office of Acquisition**. Federal awards made under RFP may include additional terms that are specific to the project, including without limitation awardee information, project description, funding details, and appropriate data rights clauses.

3. Reporting.

Applicants are advised to review 2 CFR Part 200, Uniform Administrative Requirements, Costs Principles, and Audit Requirements for Federal Awards, as supplemented by 2 CFR Part 1201, for applicable reporting requirements. Federal awards made under this RFP may include additional reporting requirements that are appropriate for the specific project to be funded by MARAD.

H. FEDERAL AWARDING AGENCY CONTACT

For any questions relating to the technical or programmatic aspects of this RFP, please contact **April Lloyd**, at April.Lloyd@dot.gov.

For any questions about the administration of this funding opportunity and any awards, please contact Christian.onwudiegwu@dot.gov and **Kelly Mitchell-Carroll** at K.mitchell-carroll@dot.gov.

I. OTHER INFORMATION

Applicants must identify all proprietary information in its submission. MARAD is not obligated to make any Federal award as a result of this RFP.